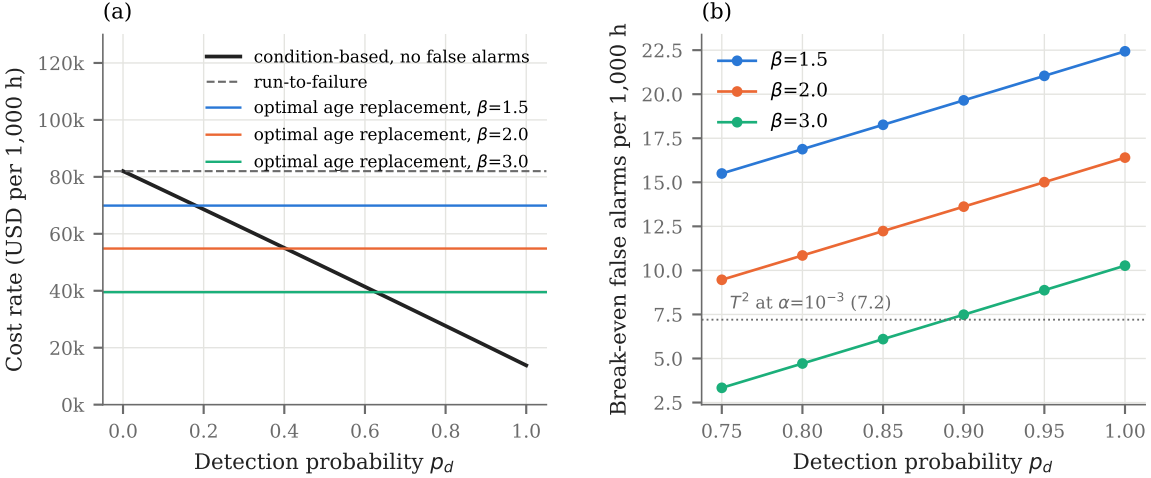
(a)
condition-based, no false alarms
run-to-failure
optimal age replacement, β=1.5
optimal age replacement, β=2.0
optimal age replacement, β=3.0
Cost rate (USD per 1,000 h)
120k
100k
80k
60k
40k
20k
0k
0.0
0.2
0.4
0.6
0.8
1.0
Detection probability p_d
(b)
β=1.5
β=2.0
β=3.0
T^2 at $\alpha=10^{-3}$ (7.2)
Break-even false alarms per 1,000 h
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.75
0.80
0.85
0.90
0.95
1.00
Detection probability p_d